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Do we still need pangenome graphs?

Seminar with Andrea Guarracino, Assistant Professor at the Translational Genomics Research Institute (TGen)

Abstract

Standard genomic analysis aligns reads to a single linear reference, which introduces bias and misrepresents complex variation. Pangenome graphs address this by jointly encoding variation across many genomes, but building and storing explicit graphs at population scale is computationally prohibitive. In this seminar I will describe an alternative: rather than constructing the full graph, we use pangenome alignment and indexed representations to answer the same queries directly. This cuts compute and memory by orders of magnitude, making population-scale analysis tractable on a single laptop.

Short Bio

Andrea Guarracino is an Assistant Professor at the Translational Genomics Research Institute (TGen) in Phoenix, Arizona. He is a core developer of PGGB and ODGI and works on pangenome alignment and representation, cancer pangenomics, and non-homologous recombination. His research aims to move genomic references from linear sequences toward population-scale pangenomes.

June 19th
h 11 am
Conference Room
Palazzo Boyl
Via Santa Cecilia 24, Pisa

More info